

QUEUE

PROGRAM CODING:

```
#include<stdio.h>

#include<conio.h>

#include<stdlib.h>

void main()

{

    int ch,n,queue[50],rear=0,front=0,i,item;

    clrscr();

    printf("\n Implementation of queue using array");

    do

    {

        printf("\n Main menu");

        printf("\n1.queue max");

        printf("\n2.add");

        printf("\n3.delete");

        printf("\n4.display");

        printf("\n5.queue size");

        printf("\n6.exit");

        printf("\n Enter your choice");

        scanf("%d",&ch);

        switch(ch)

        {

            case 1:
```

```

{
    printf("\n Enter the queue size");
    scanf("%d",&n);
    break;
}
case 2:
{
    if((front==n)|| (rear>n))
        printf("\n The queue is full");
    else
    {
        printf("\n Enter the element");
        scanf("%d",&item);
        rear=rear+1;
        queue[rear]=item;
        printf("\n The element is added");
    }
    break;
}
case 3:
{
    if(rear==front)
    {
        printf("\n The queue is empty");
    }
    else
    {

```

```

        front=front+1;

        item=queue[front];

        printf("\n The deleted element is %d",item);

    }

    break;
}

case 4:

{
    if(rear==front)

        printf("\n The queue contains");

        for(i=front+1;i<=rear;i++);

        {

            printf("\t %d",queue[i]);

        }

    break;

}

case 5:

{
    if(rear==front)

        printf("\n The queue is empty");

    else

        printf("\n The size of queue is %d",(rear-front));

}

case 6:

{

    exit(1);

}

```

```
    }  
    }while(ch<7);  
    getch();  
}
```